

1NF (First Normal Form)

Rule

- * No repeating groups
- * Each column contains a single value (atomic)

✗ Not 1NF

StudentID	Name	Subjects
1	John	Math, Science

`Subjects` contains multiple values.

✓ 1NF

StudentID	Name	Subject
1	John	Math
1	John	Science

2NF (Second Normal Form)

Rule

- * Must be in 1NF
- * No partial dependency

A non-key column should depend on the whole composite key, not part of it.

Table:

StudentID	Subject	StudentName
1	Math	John
1	Science	John

Primary Key = (StudentID, Subject)

Problem:

`StudentName` depends only on `StudentID`, not on the full key
`(StudentID, Subject)`.

This is a partial dependency.

✅ 2NF

Students

StudentID	StudentName
1	John

StudentSubjects

StudentID	Subject
1	Math
1	Science

3NF (Third Normal Form)

Rule

- * Must be in 2NF
- * No transitive dependency

Example:

StudentID	DepartmentID	DepartmentName
1	D1	Computer Science

Primary Key = StudentID

Problem:

StudentID → DepartmentID

DepartmentID → DepartmentName

So:

StudentID → DepartmentName

through another column.

This is a transitive dependency.

✅ 3NF

Students

StudentID	DepartmentID
1	D1

Departments

DepartmentID	DepartmentName
D1	Computer Science

BCNF (Stronger than 3NF)

Rule

Every determinant must be a candidate key.

Example:

| Teacher | Subject |

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Alice	Math
Bob	Science

Assume:

Teacher → Subject
 Subject → Teacher

Both are candidate keys.

BCNF satisfied.

Example violating BCNF:

Student	Course	Instructor
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John	SQL	Mike
Alice	SQL	Mike

Rules:

(Student, Course) → Instructor
Course → Instructor

`Course` is not a candidate key.

Therefore BCNF violation.

Split:

Courses

Course	Instructor
SQL	Mike

StudentCourses

Student	Course
John	SQL
Alice	SQL

4NF (Fourth Normal Form)

Rule

- * Must be BCNF
- * No Multi-Valued Dependency (MVD)

Example:

A student can have multiple hobbies and multiple languages.

Student	Hobby	Language
John	Football	English
John	Football	Tamil
John	Chess	English
John	Chess	Tamil

Problem:

Hobbies and languages are independent.

Student →→ Hobby
Student →→ Language

This is a Multi-Valued Dependency.

Why problem?

If John learns a new language:

Tamil, English, Hindi

Need to create:

Student	Hobby	Language
-----	-----	-----
John	Football	Hindi
John	Chess	Hindi

Many unnecessary rows.

✓ 4NF

StudentHobbies

Student	Hobby
-----	-----
John	Football
John	Chess

StudentLanguages

Student	Language
-----	-----
John	English
John	Tamil

Now adding Hindi needs only one row.

5NF (Fifth Normal Form)

Rule

- * Must be in 4NF
- * No Join Dependency

Table:

Supplier	Product	Project
S1	P1	PR1
S1	P2	PR1
S2	P1	PR1

Suppose business rules are:

- * Supplier can supply Product
- * Supplier works on Project
- * Product used in Project

These can be stored separately.

SupplierProduct

Supplier	Product
S1	P1
S1	P2
S2	P1

SupplierProject

Supplier	Project
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S1	PR1
S2	PR1

ProductProject

Product	Project
-----	-----
P1	PR1
P2	PR1

If the original table can always be reconstructed by joining these tables, then decomposing removes the join dependency and achieves 5NF.

Conclusion

NF	Removes
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1NF	Repeating groups
2NF	Partial dependency
3NF	Transitive dependency
BCNF	Determinant not candidate key
4NF	Multi-valued dependency

| 5NF | Join dependency |

Think of it as:

1NF → duplicate values

2NF → dependency on part of key

3NF → dependency through another column

4NF → multiple independent lists

5NF → unnecessary joins between multiple relationships